13.4.5 Wrap-Up: Error Analysis

Tangela was asked to circle all of the expressions equivalent to $6(y+1)$.

Her work is shown below.

$6y+1$	$6y+7$	$6(y)+1(y)$	$6(y)+6(1)$	$6y+6$

1. Determine if Tangela correctly identified all of the equivalent expressions. If she is missing an expression, circle the expression(s) she is missing.

2. If Tangela made an error, then explain her error and list the equivalent expressions.

Unit 13, Lesson 5: Writing Algebraic Expressions

Benchmark

MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.

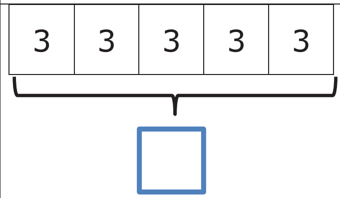
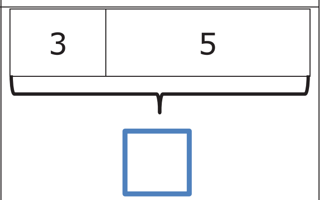
Learning Targets

- ☐ I can translate a written description into an algebraic expression.
- ☐ I can translate a real-world description into an algebraic expression.

13.5.1 Warm-Up: Bell Work

1. Two diagrams are shown. One represents $3 + 5 = 8$. The other represents $5 \cdot 3 = 15$.

   Complete the table by writing the equation that matches each diagram in the space provided and labeling the length of each diagram.

Equation		
Diagram		

2. Draw a diagram that represents each equation.

   a. $4 + 3 = 7$

   b. $4 \cdot 3 = 12$

13.5.2 Guided Instruction: Making Sense with Models

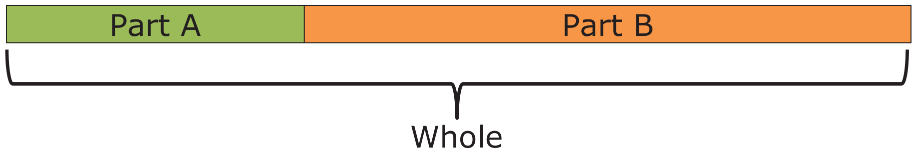
An ***algebraic expression*** is a mathematical statement containing numbers, operators, and at least one unknown, represented by *a* variable.

1. Complete the statement.

An expression ○ does ○ does not have an equal sign or inequality symbol.

Bar models help connect the action of a problem to an algebraic expression.

Part-Part-Whole Model



The part-part-whole model can be used to show the following actions.

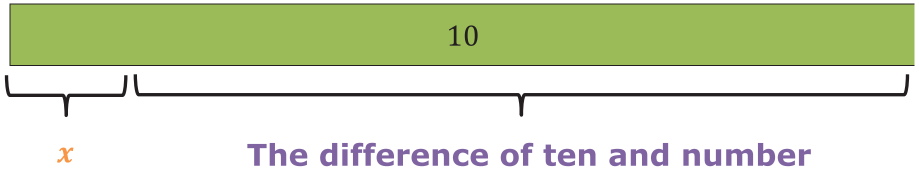
Action	Equation
Combining or joining	Part A + Part B = Whole
Taking away or separating	Whole − Part A = Part B Whole − Part B = Part A

2. "The sum of a number and three" can be shown as the length of the whole bar where one part is n and the other part is 3.

The sum of a number and three

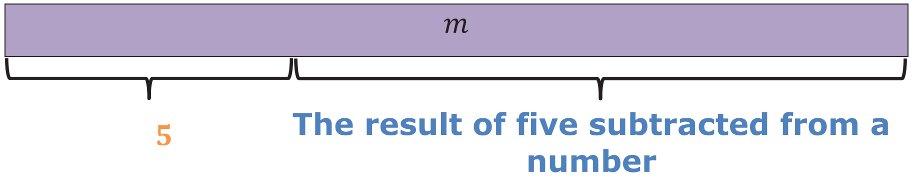
Write an algebraic expression for "The sum of a number and three."

3. "The difference of ten and a number" can be shown where the whole bar is 10 and one part is a number represented by a variable, x. The difference is the other part.



Write an algebraic expression for "The difference of ten and a number."

4. "Five subtracted from a number" is shown with the whole bar as a number represented by a variable, m, and one part is 5. The other part is the result of 5 being subtracted from the number, m.

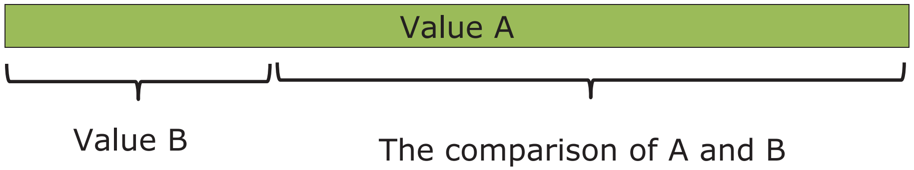


Complete the statement.

The algebraic expression for "Five subtracted from a number, m" is ________________.

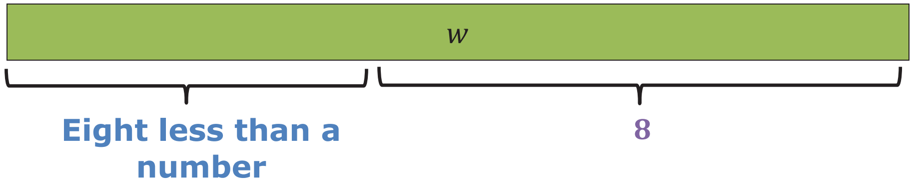
Addition/Subtraction Comparison Model

Comparison models are much like part-part-whole models. They show how much more or less one value is than another.



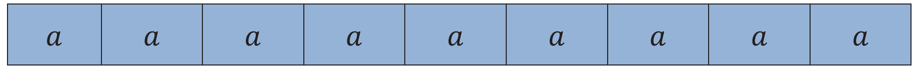
How much **more** Value A is **than** Value B, or
How much **less** Value B is **than** Value A

5. Write an algebraic expression for the bar model shown.



Algebraic expression: ____________________

6. "The product of nine and a number" can be shown as 9 equal parts with size of "a number" represented with the variable, a.

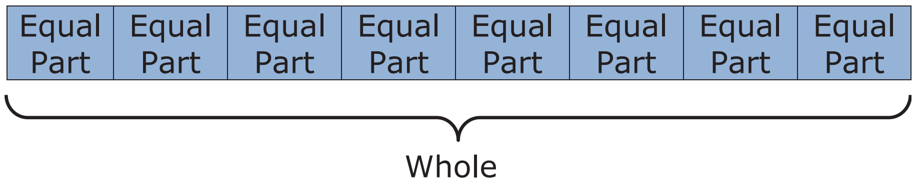


The product of nine and a number

Write an algebraic expression for "the product of nine and a number."

Equal Parts – Whole Model

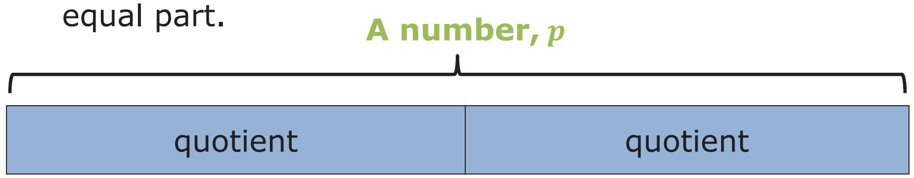
The Equal Parts – Whole Model shows the whole sectioned into equal parts.



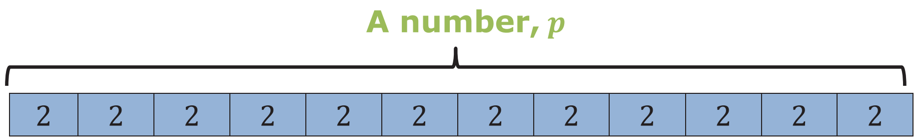
The equal parts-whole model can be used to show the following actions.

Action	Equation
Repeated addition (multiplication)	Equal part + … + equal part = whole Equal part × number of parts = whole
Equal sharing	Whole ÷ number of parts = equal part
Partitioning	Whole ÷ equal part = number of parts

7. "The quotient of a number and two" could be shown as "a number" represented by the variable, p, sectioned off into two equal parts. The quotient is the size of each equal part.



"The quotient of a number and two" could also be shown as "a number" represented by the variable p, sectioned off into equal parts with size of 2 units. The quotient is the number of equal parts.

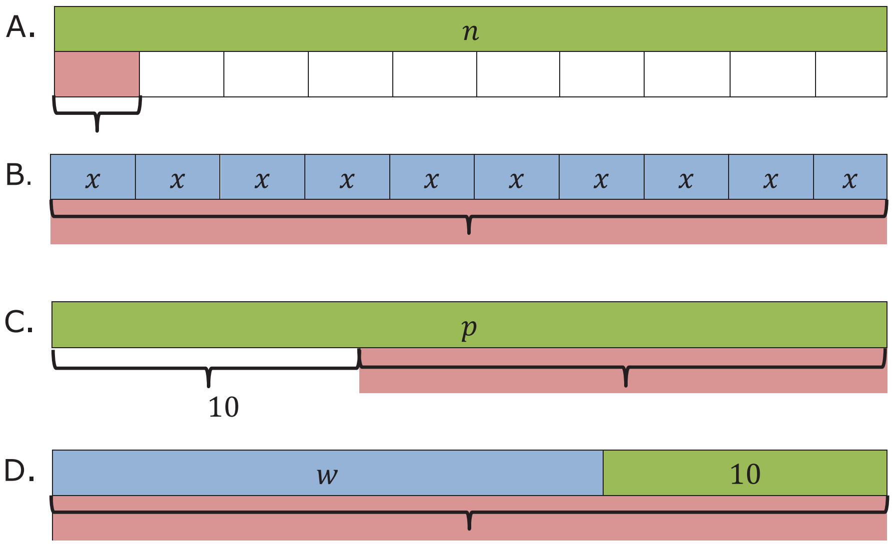


What is the algebraic expression for "The quotient of a number and 2?"

13.5.3 Collaboration: Matching Activity

Part 1
Match each written description to the appropriate bar model, where the red portion is the result of the written description. Write the letter of the bar model beside the corresponding word description.

Written Description	Bar Model
1. Ten subtracted from a number	
2. A number divided by ten	
3. The sum of a number and ten	
4. A number times ten	



Part 2
Turn and talk with your partner to compare your matches. Then, work with your partner to write an algebraic expression that represents the written description and bar model.

Written Description	Algebraic Expression
1. Ten subtracted from a number	
2. A number divided by ten	
3. The sum of a number and ten	
4. A number times ten	

13.5.4 Guided Instruction: Writing Algebraic Expressions From Descriptions of Real-World Contexts

Algebraic expressions can be used to represent real-world situations.

1. Misty has 34 pairs of shoes arranged on a number of shelves. The same number of shoes are on each shelf. She has h number of shelves.

    a. Write an algebraic expression that represents this situation.

    b. What does the algebraic expression represent?

2. Martin buys 8 pounds of apples at a price per pound to make pie. Each pound costs p dollars.

    a. Write an algebraic expression to represent this situation.

    b. What does the algebraic expression represent?

3. Paige is 6 inches taller than her brother. Paige's brother is m inches tall.

    a. Write an algebraic expression to represent this situation.

    b. What does the algebraic expression represent?

13.5.5 Try It: Writing Algebraic Expressions

Write an algebraic expression to represent each situation.

1. Joe is 9 years younger than Drew, who is n years old.

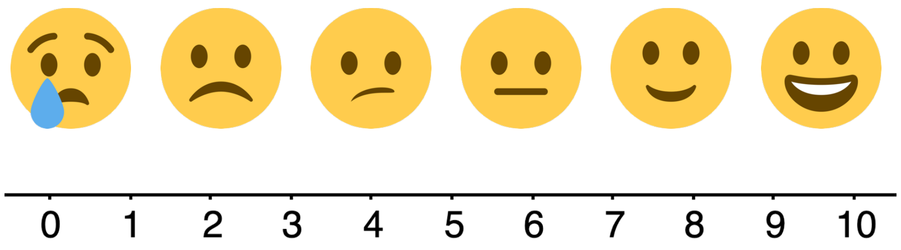
2. Jenna's salary, s, is lowered by $\$120$.

3. Maria has 12 cupcakes and wants to split them among f friends.

4. Target has 8 cookie packages with c cookies in each.

13.5.6 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of translating a written description into an algebraic expression.



2. Complete the statement.

I chose this emoji because ______________________________

____

____